Impact energy and TNT equivalents

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Convert an energy estimate without claiming a complete damage prediction.

SUCCESS CRITERIA - I can...

1. Convert speed units before squaring.
2. Calculate kinetic energy from stated model values.
3. Distinguish an energy equivalent from an explosion or damage forecast.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “equivalent” and “kinetic energy” to explain your model or result. Sentence starters are optional.

equivalent: equal in the specified quantity, not necessarily in other effects

kinetic energy: energy associated with motion

impact: a collision or strike between objects

01 Warm-up: bring back an earlier idea

A constant 30 N forward force acts at steady 5 m/s for 7 s. **Find** mechanical power and work.

Word help: power = rate of energy transfer

Sentence starter: I found ... [unit]. This value means ... in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $K = \frac{1}{2}mv^2$
TNT equivalent (tons) = $K/(4.184\times10^9\ J)$.
1 ton TNT equivalent = 4.184×10^9 J. Equal energy does not imply identical effects.

Worked reasoning: A fictional event transfers 8.4×10^6 J. Using 4.2×10^6 J per kg TNT as an energy comparison gives 2 kg TNT equivalent. This compares energy, not identical effects.

Your turn: Draw a labeled energy comparison and name one effect this number alone cannot predict.

Sentence starter: I predict ... . My model supports this because ... .

Initial **Transfers** **Final**

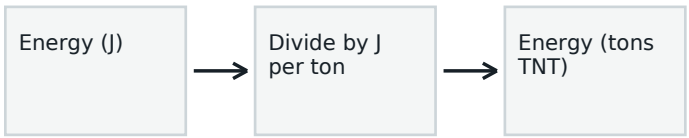
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

03 Read and interpret the model

Two reports describe the same fictional impact: one uses joules and the other uses TNT-equivalent tons.
Use the conversion model to explain why the units can express the same quantity. What does this comparison leave out?

Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

04 Apply the physics

A hypothetical 10^6 kg asteroid moves at 10 km/s.
Convert speed to m/s, calculate kinetic energy and express that energy in TNT-equivalent tons using 4.184×10^9 J per ton. Complete the conversion table.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Energy (J)	J per ton TNT	Equivalent tons
___	4.184×10^9	___

Marzano check - circle: 1 2 3

05 Explain your reasoning

Why should an energy-equivalent number not be reported as a precise area of destruction?

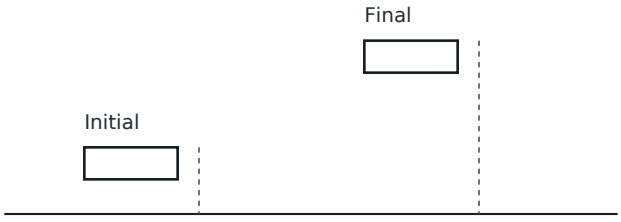
Sentence starter: Under the assumption ..., the evidence ... supports ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 5 kg object descends 7 m near Earth.
Use g=10.
Find ΔU and the work done by gravity.
Label the distances or energy levels used in your reasoning.

Sentence starter: I found ... [unit]. This value means ... in this situation.



Choose and label a zero-height level.

Marzano check - circle: 1 2 3

07 Exit check

At fixed speed, mass doubles while assumed energy coupling fraction halves.
Compare coupled energy.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both ... . They differ in ... because ... .

Initial **Transfers** **Final**



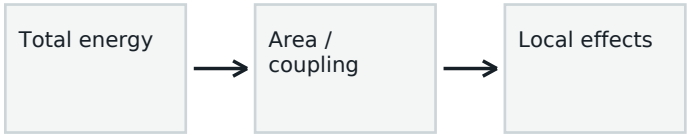
Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

A 2000 kg hypothetical object moves at 1000 m/s.
Find kinetic energy and kg TNT equivalent using 4.2×10^6 J/kg.
Explain why neither alone predicts a crater size.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both ... . They differ in ... because ... .



Marzano check - circle: 1 2 3